Thermodynamic State Vector Interface Contracts and Non-Equilibrium Entropy Generation in the Web of Life Biosphere Framework

Lead Scientific Communications & Academic Outreach Agent
Web of Life Research Consortium
https://github.com/pascalranoroarijaona/WebOfLife

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Abstract

We present the formalization and TypeScript implementation of thermodynamic state vector interface contracts, entropy generation rate ($\dot{S}_{\text{gen}}$) quantification, and exergy destruction tracking ($\dot{I} = T_0\dot{S}_{\text{gen}}$) for the Web of Life biosphere simulation framework (Sprint 25). Within complex systems ecology, maintaining physical admissibility across biochemical cycles (Carbon, Nitrogen, Phosphorus, and Water) requires strict adherence to the First and Second Laws of Thermodynamics. By wrapping state transitions in immutable monad structures that enforce non-negative internal entropy generation ($\dot{S}_{\text{gen}} \geq 0$), our architecture prevents non-physical backflow of entropy and guarantees rigorous geophysical validity under solar radiative forcing.

Keywords: Non-Equilibrium Thermodynamics, Second Law of Thermodynamics, Entropy Generation Rate, Exergy Destruction, Systems Ecology, Biosphere Simulation.

1 Introduction and Systems Ecology Foundations

The Web of Life simulation engine models global biogeochemical cycles as open thermodynamic systems driven exclusively by external solar radiation and bounded by terrestrial thermal dissipation. To prevent numerical drift and physical violations within simulated ecosystems, internal state mutations must obey foundational thermodynamic conservation and dissipation laws.

Sprint 25 establishes strict TypeScript interface contracts within `src/thermodynamics/types.ts` and executable monad methods within `src/thermodynamics/methods.ts`. These modules provide rigorous mathematical scaffolding for tracking energy, matter, and entropy across spatial-temporal boundaries.

2 Thermodynamic Governing Equations

2.1 First Law: Conservation of Energy and Matter

The evolution of total system energy adheres to the open-system energy balance:

$$\frac{dE_{\text{system}}}{dt} = \sum \dot{Q}_k - \dot{W}_{\text{system}} + \sum_i \dot{m}_i \left(h_i + \frac{v_i^2}{2} + gz_i \right) \tag{1}$$

Mass fluxes across boundaries are strictly tracked such that $\Delta M_{\text{system}} = \sum \int (\dot{m}_{\text{in}} - \dot{m}_{\text{out}})dt$.

2.2 Second Law: Entropy Balance and Generation

Local entropy changes are driven by external heat exchanges across boundary layers plus internal irreversible dissipations:

$$\frac{dS_{\text{system}}}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \dot{S}_{\text{gen}} \tag{2}$$

The internal entropy generation rate $\dot{S}_{\text{gen}}$ is constrained by the Second Law to be non-negative:

$$\dot{S}_{\text{gen}} = \dot{S}_{\text{thermal}} + \dot{S}_{\text{chemical}} + \dot{S}_{\text{diffusive}} \geq 0 \tag{3}$$

2.3 Gouy-Stodola Theorem: Exergy Destruction

Exergy destruction ($\dot{I}$) is quantified directly via the Gouy-Stodola theorem, utilizing an ambient reference temperature $T_0 = 298.15$ K:

$$\dot{I} = T_0 \cdot \dot{S}_{\text{gen}} \tag{4}$$

3 Architecture and Implementation

The core TypeScript contracts and monad transition wrappers enforce these laws at every simulation step. Every biogeochemical state transition passes through `ThermodynamicMonad`, which evaluates $\dot{S}_{\text{gen}}$ and immediately throws a `ThermodynamicViolationError` if $\dot{S}_{\text{gen}} < 0$.

4 Summary of Biogeochemical Cycle Constraints

Process / Cycle	Mass Delta (Δm)	Energy Delta (ΔE)	Entropy Gen ($\dot{S}_{\text{gen}}$)	Exergy Dest ($\dot{I}$)
Solar Forcing	0 (Radiative)	$+\dot{Q}_{\text{solar}}$	≥ 0	$T_0 \dot{S}_{\text{gen, solar}}$
Carbon Cycle	$\sum \Delta m_{\text{in}} = \sum \Delta m_{\text{out}}$	$+\Delta H_{\text{reaction}}$	≥ 0	$T_0 \dot{S}_{\text{gen, photo}}$
Nitrogen Fixation	0 (Closed mass)	$+\Delta H_{\text{ATP}}$	≥ 0	$T_0 \dot{S}_{\text{gen, N-fix}}$
Hydrological Cycle	$\Delta m_{\text{liquid}} = -\Delta m_{\text{vapor}}$	$+\Delta H_{\text{latent}}$	≥ 0	$T_0 \dot{S}_{\text{gen, vapor}}$

Table 1: Thermodynamic compliance matrix across simulated biogeochemical cycles.

5 Conclusion

Sprint 25 successfully establishes the thermodynamic backbone of the Web of Life simulation engine. Future sprints will extend these interface contracts to spatial diffusion grids and macro-ecological trophic networks.

For further details, source code, and release logs, please consult the official repository: `https://github.com/pascalranoroarijaona/WebOfLife`.